About Dataset

Context

Dream Housing Finance Company deals in all kinds of home loans. They have a presence across all urban, semi-urban, and rural areas. The customer first applies for a home loan and after that company validates the customer's eligibility for a loan.

Company wants to automate the loan eligibility process (real-time) based on customer detail provided while filling online application form. These details are Gender, Marital Status, Education, Number of Dependents, Income, Loan Amount, Credit History and others. To automate this process, they have provided a dataset to identify the customer segments eligible for loan amounts to target these customers specifically.

Dataset Glossary (Column-wise)

Train file: CSV containing the customers for whom loan eligibility is known as **'Loan_Status'**

Variable	Description
Loan_ID	Unique Loan ID
Gender	Male/ Female
Married	Applicant married (Y/N)
Dependents	Number of dependents

Education	Applicant Education (Graduate/ Under Graduate)
Self_Employed	Self-employed (Y/N)
ApplicantIncome	Applicant income
CoapplicantIncome	Co-applicant income
LoanAmount	Loan amounts in thousands
Loan_Amount_Term	Term of loan in months
Credit_History	credit history meets guidelines
Property_Area	Urban/ Semi Urban/ Rural
Loan_Status	(Target) Loan approved (Y/N)

Task

1. Read the CSV file and see if there is any null value. Take necessary actions(for example: imputation of missing values) to deal with the rows that contain null values. (5)
2. Make a pair plot, using 'Loan_Status'' as the hue. (2)
3. Drop the columns which does not affect the target column. (2)
4. Split the dataset. Keep 20% of the data for testing purposes. (2)
5. Apply encoding if necessary. (3)
6. Apply feature scaling if necessary. (2)
7. Use a logistic regression model on X_train. (3)
8. Use Grid Search CV to find the best hyperparameters. **Use the F1 score as the scoring criterion**. (5)
9. Predict using your model and print the F1 score. (2)
10. Print the confusion matrix and explain the result in a text box. (3)